

Randomized reaching: every waypoint pair, both arms independent

Question: when the two arms are given *unrelated* targets, how often can the pair serve them at once? **Test:** all $450 \times 449 = 202,050$ **ordered waypoint pairs** per layout — left arm to one point, right arm to another — **606,150 configurations** in total.

Produced by `randomized_reaching.py` (56 workers, ~7 min per layout); verdict codes in `randomized_reaching.npz`. Same pass rules as `bimanual_test.py`: both arms solve IK, touch nothing during the approach, and settle parked within 20 mm.

This differs from `REACHABILITY_REPORT.md`, where both arms serve the *same* waypoint offset by a fixed separation. Here nothing coordinates them, which is what exposes interference.

Headline

verdict	urgantry	urtable_45	urtable
PASS	87,039 (43.1 %)	75,202 (37.2 %)	63,294 (31.3 %)
collision — arm vs arm	83,394 (41.3 %)	93,625 (46.3 %)	131,663 (65.2 %)
collision — arm vs structure	2,616 (1.3 %)	14,113 (7.0 %)	0 (0.0 %)
droop	26,257 (13.0 %)	17,876 (8.8 %)	6,196 (3.1 %)
unstable	29 (0.0 %)	1,098 (0.5 %)	0 (0.0 %)
no IK	2,715 (1.3 %)	136 (0.1 %)	897 (0.4 %)

Most configurations fail in every layout, and the dominant cause is always the arms hitting each other. That is a property of random pairing, not a defect: a large share of pairs asks the two arms to swap sides of the table.

The ordering matches the shared-target test (`urgantry` $\geq$ `urtable_45` $>$ `urtable`), but the margins are far wider — 43.1 % vs 31.3 % here, against 84 % vs 76 % there. Independent targets separate the layouts much more sharply than a shared one.

The result that matters: crossing

Splitting by where each arm is sent explains nearly everything:

	urgantry	urtable_45	urtable	n
uncrossed (left arm $\rightarrow -x$, right arm $\rightarrow +x$)	91.6 %	79.9 %	86.9 %	44,100
same side (both targets same half)	37.4 %	30.4 %	18.8 %	88,650
crossed (left arm $\rightarrow +x$, right arm $\rightarrow -x$)	3.0 %	3.1 %	0.7 %	44,100

Arm-vs-arm collision rate over the same split:

	urgantry	urtable_45	urtable
uncrossed	0.2 %	2.0 %	7.5 %
same side	38.4 %	48.3 %	77.0 %
crossed	90.0 %	89.5 %	98.3 %

Crossing is fatal in every layout. When the arms are asked to swap sides they collide in ~90 % of configurations, and `urtable` — with 20 cm between its bases — fails 98.3 % of them. No layout tested here tolerates crossed assignment.

Uncrossed assignment is the opposite story. Give each arm the side it is on and `urgantry` serves 91.6 % of pairs and `urtable` 86.9 %. The headline 31–43 % figures are dominated by the 44,100 crossed pairs and the 88,650 same-side pairs that a real task allocator would never generate.

The practical reading: **the layout matters much less than the assignment policy.** A scheduler that assigns each target to the nearer arm converts a ~40 % success rate into a ~90 % one on the same hardware — a bigger gain than any layout change on offer.

Where each layout actually differs

Once crossing is set aside, the layouts differ in *how* they fail:

- `urgantry` — best overall (43.1 %) and best uncrossed (91.6 %), with almost no structural collisions (1.3 %). Its distinctive failure is **droop**: 13.0 %, four times `urtable`'s rate, from arms hanging at long reach.
- `urtable_45` — the only layout with meaningful **structural** collisions (7.0 %, 14,113 configs): arms striking their own angled stand. Also the only one with `unstable` verdicts (1,098). Both are fixture problems, not kinematic ones.
- `urtable` — **zero** structural collisions and the lowest droop (3.1 %), because its upright bases and short reaches are mechanically easy. It pays for that with the highest arm-vs-arm rate in every geometry class, including 7.5 % even when *uncrossed*, where the other two are at 0.2 % and 2.0 %.

That last number is the sharpest single indictment of the 20 cm base spacing: `urtable` is the only layout that collides appreciably even when each arm stays on its own side.

Target separation

PASS rate against the distance between the two arms' targets:

separation	urgantry	urtable_45	urtable	n
0.00–0.15 m	17.3 %	18.4 %	0.0 %	8,778
0.15–0.30 m	39.9 %	33.5 %	14.4 %	56,296
0.30–0.45 m	47.6 %	41.6 %	37.0 %	73,674
0.45–0.60 m	44.0 %	39.2 %	44.7 %	44,778
0.60–0.80 m	45.1 %	35.5 %	42.9 %	18,216

urtable cannot serve **a single one** of the 8,778 pairs whose targets are within 15 cm of each other — its two grippers cannot occupy adjacent points at all. The other two manage ~18 %. This is the quantitative version of the 0.35 m **separation** constant used in the shared-target test: below that spacing **urtable** simply does not function bimanually.

Beyond ~0.45 m all three plateau, and **urtable** catches up — at wide separation the arms are far enough apart that only reach matters.

Layer pairs

PASS rate by the height layer of each arm's target:

left / right	urgantry	urtable_45	urtable
low / low	37.0 %	26.7 %	23.3 %
mid / mid	44.0 %	35.8 %	20.9 %
high / high	41.6 %	46.9 %	20.7 %
low / high (either order)	~45 %	~38 %	~42 %

urtable_45 is the only layout that improves with height (26.7 % → 46.9 %), consistent with its angled stand lifting the arms clear of the board. **urtable** degrades at height (23.3 % → 20.7 %) because reaching up brings both arms into the same volume above their closely-spaced bases. Mixed-height pairs are easiest for every layout — one arm high and one low keeps them out of each other's way.

Recommendation

1. **Fix the assignment policy before the layout.** Nearest-arm assignment moves success from ~40 % to ~90 % on unchanged hardware. Crossing is unrecoverable in all three layouts.
2. **urgantry is the most robust to uncoordinated targets** — best overall, best uncrossed, and its failures are droop, which a stiffer controller or a looser tolerance addresses.
3. **urtable should not be used for genuinely independent bimanual work.** 65 % arm-vs-arm collisions, 7.5 % even uncrossed, and total failure below 15 cm target separation.

4. `urtable_45`'s **7 % structural collisions are the cheapest fix in this report** — reshape the pedestal and it likely overtakes `urgantry` on uncrossed work.
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Method

- All ordered pairs $(i, j), i \neq j$, of the 450-waypoint grid. Ordered, not unordered: the arms are distinct, so left→A/right→B differs from left→B/right→A.
- Each configuration restarts from the home pose, both arms IK to their own target tool-down, 4 s approach with contact monitoring every step, then a 1 s hold.
- Verdict precedence: `no IK` → `collision` (split arm-vs-arm / arm-vs-structure by whether both contact sides belong to arms) → `unstable` → `droop` → `drifted` → `PASS`.
- Props removed; `implicitfast` integrator (Euler chatters against these stiff servos and no pose reads as settled).
- 56 worker processes, one BLAS thread each, ~500–600 configs/s aggregate, ~7 min per layout. Chunks are retried with a smaller pool if a worker dies — MuJoCo took a process down hard twice during these runs, at different configurations each time.
- `error` verdicts (diverged physics or a crashed chunk) came out at **0** for all three layouts, so no result here rests on a failed configuration.

Caveats

- **A `collision` verdict is evidence, not proof.** The IK is damped least squares warm-started from home, taking the first converging random restart on failure. It does not enumerate the ~8 analytic UR branches or prefer least motion, and there is no path planning — joints interpolate independently. A planner could clear some of these collisions, particularly in the same-side class. The crossed-pair result is unlikely to move: the arms must physically pass through each other's volume.
- **No task semantics.** Every pair is weighted equally, including pairs no sensible allocator would produce. The conditional tables, not the headline, are the usable result.
- **Static targets only** — no payload, no motion through the pair, no handovers.
- Simulation: no friction model, no gearbox compliance, no backlash.